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(54) **USER EQUIPMENT BEHAVIOR AND REQUIREMENTS FOR POSITIONING MEASUREMENT WITHOUT GAP**

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H04W 24/10 (2009.01)
H04W 64/00 (2009.01)

(52) **U.S. Cl.**
CPC ***H04L 5/0051*** (2013.01); ***H04W 24/10*** (2013.01); ***H04W 64/00*** (2013.01)

(58) **Field of Classification Search**
CPC H04L 5/0051; H04W 24/10; H04W 64/00
See application file for complete search history.

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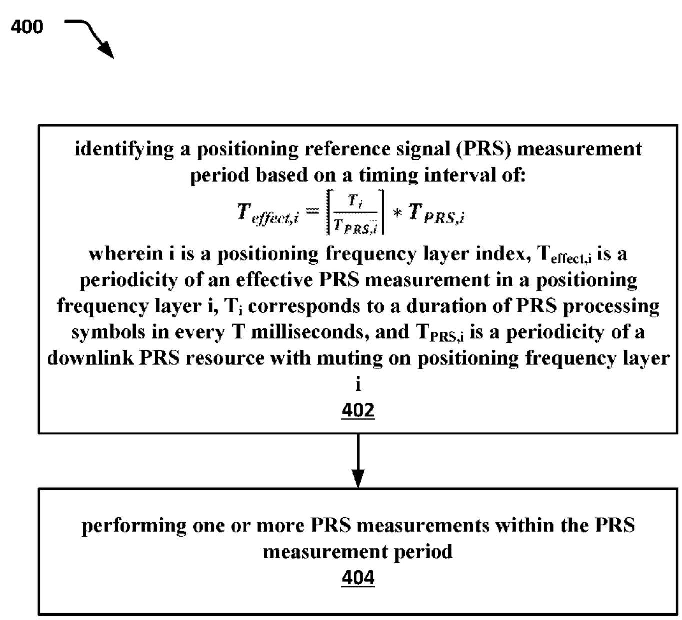
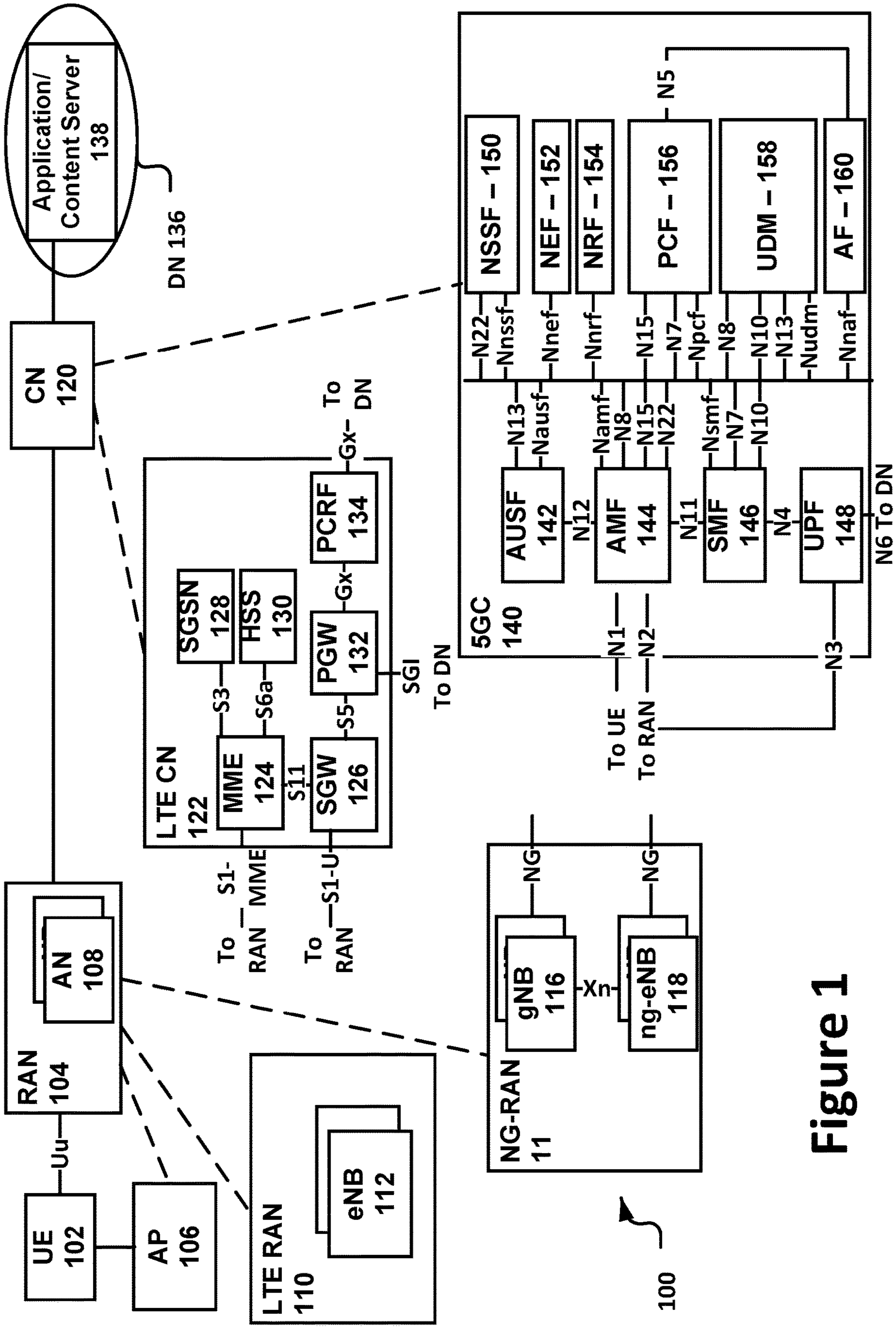
(57) **ABSTRACT**

Various embodiments herein provide techniques for positioning measurements in a wireless cellular network without a measurement gap. The measurements may be performed in one or more positioning frequency layers (PFLs). Additionally, techniques for activation and/or deactivation of a measurement gap are provided. Other embodiments may be described and claimed.

20 Claims, 5 Drawing Sheets

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DN 136
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MME 124
SGW 126
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HSS 130
PGW 132
PCRF 134
S1-MME
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To RAN
S3
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S11
Gx
SGI To DN
To DN
5GC 140
AUSF 142
AMF 144
SMF 146
UPF 148
N12
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N4
N6 To DN
N1
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To UE
To RAN
NSSF – 150
NEF – 152
NRF – 154
PCF – 156
UDM – 158
AF – 160
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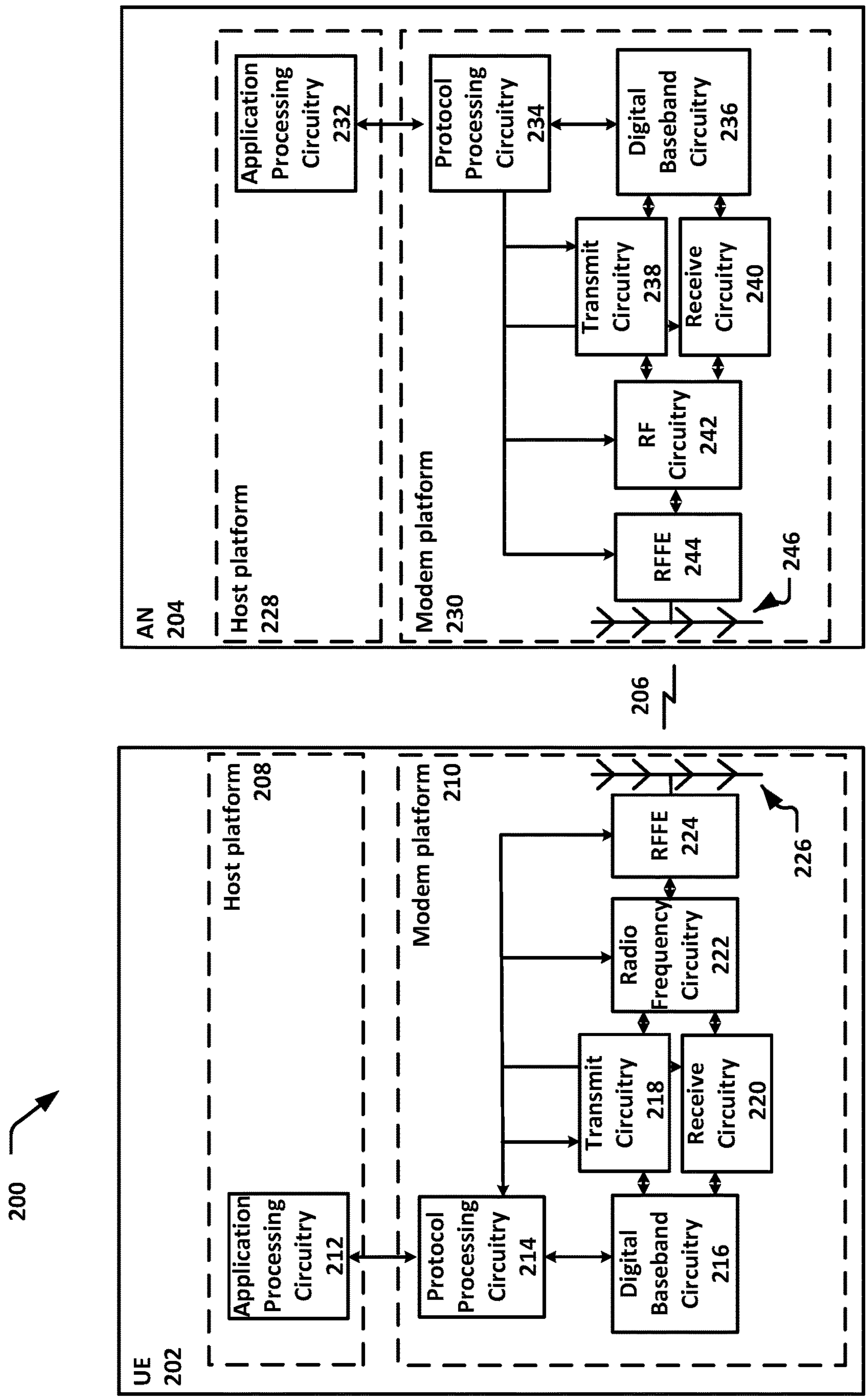
Figure 1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Host platform
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Application Processing Circuitry 212
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Protocol Processing Circuitry 214
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Transmit Circuitry 218
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Radio Frequency Circuitry 222
RFFE 224
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Host platform
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Application Processing Circuitry 232
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Protocol Processing Circuitry 234
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Transmit Circuitry 238
Receive Circuitry 240
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RFFE 244
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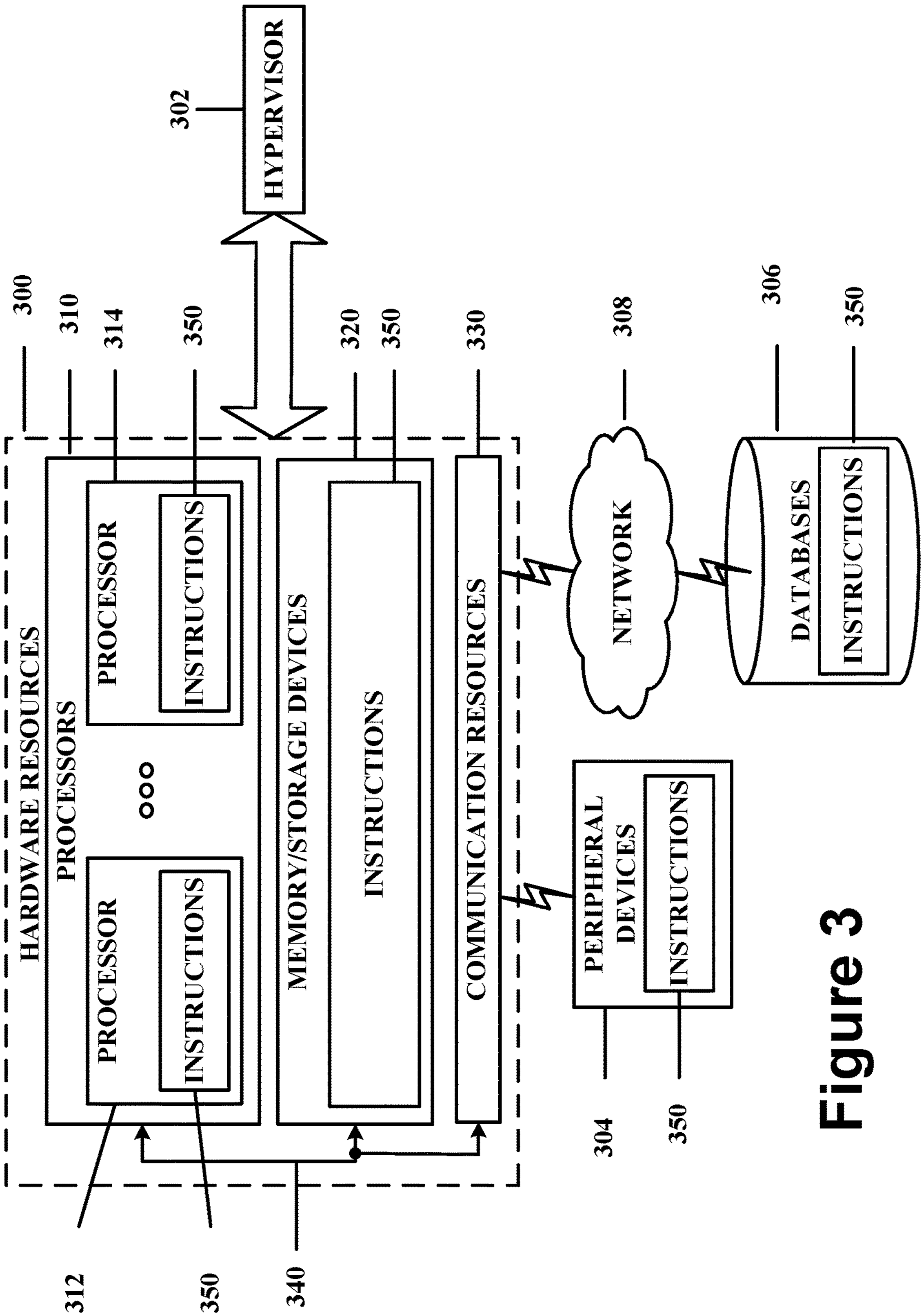
Figure 2


HYPERVISOR
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INSTRUCTIONS
MEMORY/STORAGE DEVICES
INSTRUCTIONS
COMMUNICATION RESOURCES
NETWORK
DATABASES
INSTRUCTIONS
PERIPHERAL DEVICES
INSTRUCTIONS
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312
350
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Figure 3

400

identifying a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i}$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i

402

performing one or more PRS measurements within the PRS measurement period

404

Figure 4

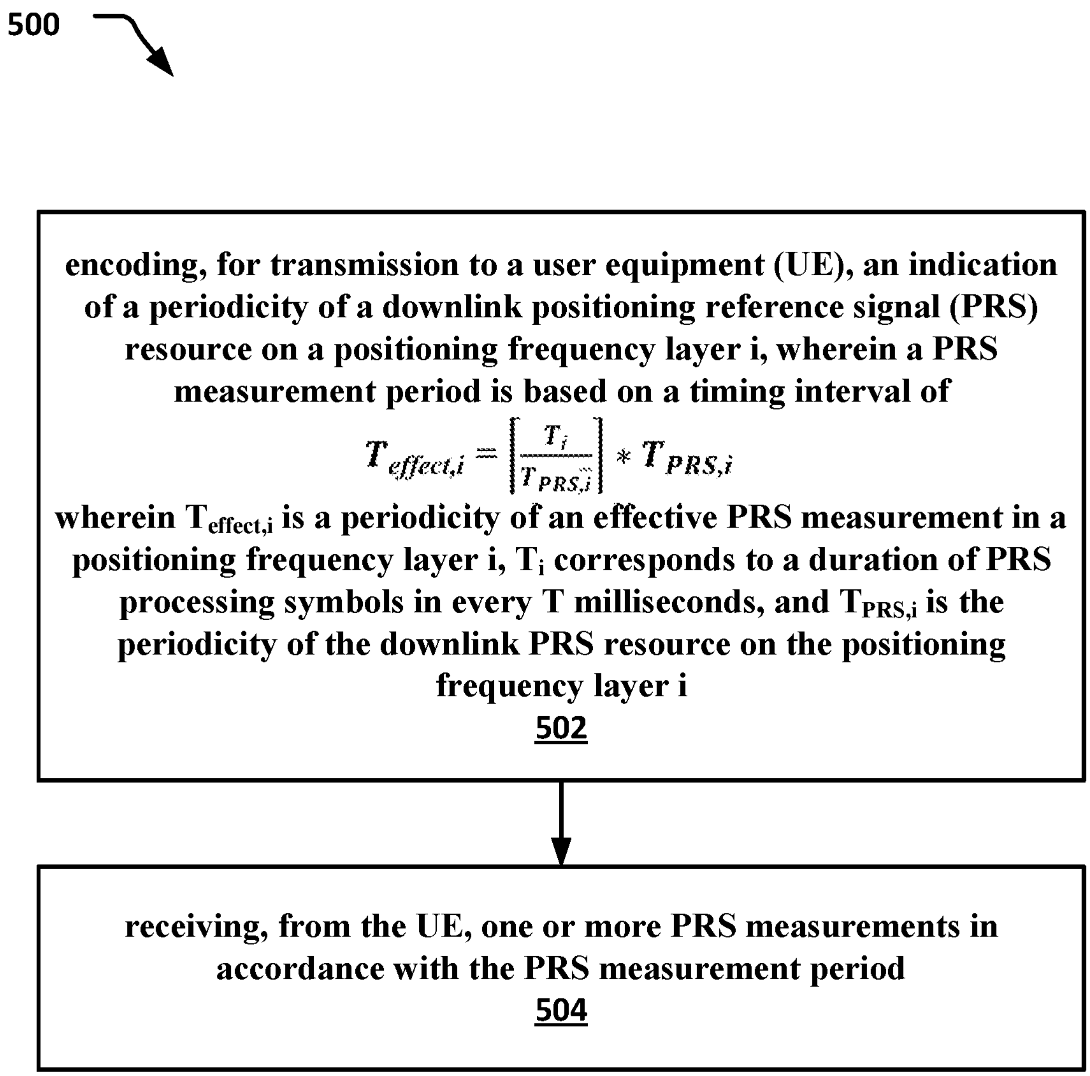
500
encoding, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of
$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i}$
wherein $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is the periodicity of the downlink PRS resource on the positioning frequency layer i
502
receiving, from the UE, one or more PRS measurements in accordance with the PRS measurement period
504

Figure 5

USER EQUIPMENT BEHAVIOR AND REQUIREMENTS FOR POSITIONING MEASUREMENT WITHOUT GAP

CROSS REFERENCE TO RELATED APPLICATION

The present application claims priority to U.S. Provisional Patent Application No. 63/297,646, which was filed Jan. 7, 2022; the disclosure of which is hereby incorporated by reference.

FIELD

Various embodiments generally may relate to the field of wireless communications. For example, some embodiments may relate to user equipment (UE) behavior and requirements for positioning measurements without a gap.

BACKGROUND

A user equipment (UE) in a wireless cellular network typically receives positioning reference signals (PRSs) from one or more next generation Node Bs (gNBs). The UE performs positioning measurements on the PRSs, such as a received signal time difference (RSTD) measurement.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings.

FIG. **1** schematically illustrates a wireless network in accordance with various embodiments.

FIG. **2** schematically illustrates components of a wireless network in accordance with various embodiments.

FIG. **3** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

FIG. **4** illustrates an example process for practicing the various embodiments herein.

FIG. **5** illustrates another example process for practicing the various embodiments herein.

DETAILED DESCRIPTION

The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrases "A or B" and "A/B" mean (A), (B), or (A and B).

Various embodiments herein provide techniques for the UE to perform positioning measurements (e.g., on a positioning reference signal (PRS)) in a wireless cellular network. For example, embodiments may relate to the UE reporting the timing error group (TEG) information for NR positioning.

With respect to the reporting requirements for the PRS measurement requirements in Rel16 [TS38.133] below, the delay requirement of DL PRS RSTD will depend on number of crucial factors which will be addressed below: the number of positioning occasions and the number of consecutive subframes, periodicity of positioning occasions, measurement sampling rate, measurement bandwidth, beam management, and other network assistance information, etc.

$$T_{RSTD,Total} = \sum_{i=1}^{L} T_{RSTD,i} + (L-1) * \max(T_{effect,i})$$

Where,

i is the index of positioning frequency layer,

L is total number of positioning frequency layers, and

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i

$T_{RSTD,i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer i as specified below:

$$T_{RSTD,i} = \left(CSSF_{PRS,i} * N_{RxBeam,i} * \left\lceil \frac{N_{PRS,i}^{slot}}{N'} \right\rceil \left\lceil \frac{L_{available\_PRS,i}}{N} \right\rceil * N_{sample} - 1\right) * T_{effect,i} + T_{last},$$

where:

$N_{RxBeam,i}$ is the UE Rx beam sweeping factor. In FR1, $N_{RxBeam,i}=1$; and in FR2, $N_{RxBeam,i}=8$.

$CSSF_{PRS,i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer i as defined in clause 9.1.5.2.

N_{PRS}^{slot} is the maximum number of DL PRS resources in positioning frequency layer i configured in a slot.

$L_{available\_PRS,i}$ is the time duration of available PRS to be measured in the positioning frequency layer i, and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26].

N_{sample} is the number of PRS RSTD samples and $N_{sample}=4$.

T_{last} is the measurement duration for the last PRS RSTD sample, including the sampling time and processing time, $T_{last}=T_i+T_{available\_PRS,i}$,

$T_{effect,i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available\_PRS,i}} \right\rceil * T_{available\_PRS,i}$$

Where,

T_i corresponds to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34],

$T_{available\_PRS,i}$=LCM(T_{PRS},i, $MGRP_i$), the least common multiple between $T_{PRS,i}$ and $MGRP_i$.

$MGRP_i$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i.

$T_{PRS,i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer i.

If more than one PRS periodicities are configured in positioning frequency layer i, the least common multiple of PRS periodicities T PRS with muting among all DL PRS resource sets in the positioning frequency layer is used to derive the measurement period of that positioning frequency layer i. Where,

$T_{per}^{PRS\ with\ muting}=N_{muting}*T_{per}^{PRS}$ is the PRS periodicity with muting per PRS resource,

T_{per}^{PRS} is the periodicity of PRS resource sets given by the higher-layer parameter DL-PRS-Periodicity.

N_{muting} is the scaling factor considering PRS resource muting. If bitmap $\{b^1\}$ for higher-layer parameter DL-PRS-MutingPattern is provided, and $T_{per}^{PRS}*T_{muting}^{PRS}\leq 10240$ ms, then

$$N_{muting} = T_{muting}^{PRS} * \min\left(L, \frac{10240}{T_{per}^{PRS} * T_{muting}^{PRS}}\right);$$

otherwise, if bitmap $\{b^1\}$ is not provided or $T_{per}^{PRS}*T_{muting}^{PRS}>10240$ ms, then $N_{muting}=1$.

T_{muting}^{PRS} is the muting repetition factor given by the higher-layer parameter DL-PRS-MutingBitRepetitionFactor, and L is the size of the bitmap $\{b^1\}$.

Note: For the purpose of calculating TPRS,i, only the PRS resources fully or partially covered by the MG are considered.

{N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to durationOfPRS-ProcessingSysmbols in TS 37.355 [34] processed every T ms corresponding to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to supportedBandwidthPRS in TS 37.355 [34].

N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by maxNumOfDL-PRS-ResProcessedPerSlot specified in TS 37.355 [34].

The time $T_{RSTD,i}$ starts from the first MG instance aligned with a DL PRS resource(s) of positioning frequency layer i closest in time after both the NR-TDOA-ProvideAssistanceData message and NR-TDOA-RequestLocationInformation message are delivered from LMF to the physical layer of UE via LPP [34].

Periodicity of PRS Occasion

For the gap-less measurement, the periodicity of PRS occasion ($T_{available\_PRS,i}$) can be same as the configured PRS periodicity ($T_{PRS,i}$) by LMF because no MGRP.

The network configured T_{PRS} is broadcasted to all UEs in a cell. Thus, according to RAN1 agreements on how to define UE processing capability ("T"), it is possible that there is misalignment between the network configured T_{PRS} and UE specific T. As a result, from RAN4 perspective, the measurement delay requirements may depend on the larger interval of available PRSs. Thus max(T, T_{PRS}) may be taken as the basic timing interval for PRS measurement latency requirements.

In some embodiments, the basic timing interval to be used to define PRS measurement period may be:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i}$$

Applicable Number of Positioning Frequency Layers (PFLs)

In some embodiments, in case of the gap-less PRS measurement, there is only single PFL to be measurement. Otherwise, the gap may be needed. But it is also feasible for the UE which can support the gap less measurement for different PFLs (e.g. the center frequency of different PFLs are same but the active BWP of them not.

Accordingly, in some embodiments, the PRS gap-less measurement requirements is applicable for the single PFL only.

In other embodiments, when the multiple PFLs configured, UE can decide perform the intra-f measurement for the PFL which can contained within UE's active bandwidth part (BWP). The UE may perform measurements on other PFLs by inter-frequency measurement.

PRS Processing Window

Ti can impact the Teffect time above.

Carrier Specific Scaling Factor (CSSF)

If the gap-less PRS measurement used, the other concurrent PRS processing and the legacy RRM measurements can be possible. The existing CSSF may be updated accordingly.

Measurement Gap (MG) Activation and Deactivation Mechanisms

In RAN1, the possible measurement gap enhancement in RAN1 were discussed. Up to now, some agreements/working assumptions in RAN1 was achieved [R1-2108639]:

Subject to UE capability, support PRS measurement outside the MG, within a PRS processing window, and UE.

As the pre-configured MG for PRS was introduced in Rel17, the corresponding requirements for this new aspect may be introduced. For example, in some embodiments, requirements may be defined for the case with MG switch, e.g. by taking into account the new MG.

Systems and Implementations

FIGS. **1-3** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

FIG. **1** illustrates a network **100** in accordance with various embodiments. The network **100** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

The network **100** may include a UE **102**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **104** via an over-the-air connection. The UE **102** may be communicatively coupled with the RAN **104** by a Uu interface. The UE **102** may be, but is not limited to, a smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

In some embodiments, the network **100** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

In some embodiments, the UE **102** may additionally communicate with an AP **106** via an over-the-air connection. The AP **106** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **104**. The connection between the UE **102** and the AP **106** may be consistent with any IEEE 802.11 protocol, wherein the AP **106** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **102**, RAN **104**, and AP **106** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **102** being configured by the RAN **104** to utilize both cellular radio resources and WLAN resources.

The RAN **104** may include one or more access nodes, for example, AN **108**. AN **108** may terminate air-interface protocols for the UE **102** by providing access stratum protocols including RRC, PDCP, RLC, MAC, and L1 protocols. In this manner, the AN **108** may enable data/voice connectivity between CN **120** and the UE **102**. In some embodiments, the AN **108** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **108** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **108** may be a macrocell base station or a low power base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In embodiments in which the RAN **104** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **104** is an LTE RAN) or an Xn interface (if the RAN **104** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

The ANs of the RAN **104** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **102** with an air interface for network access. The UE **102** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **104**. For example, the UE **102** and RAN **104** may use carrier aggregation to allow the UE **102** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG and a second AN may be secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

The RAN **104** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Prior to accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

In V2X scenarios the UE **102** or AN **108** may be or act as a RSU, which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by: a UE may be referred to as a "UE-type RSU"; an eNB may be referred to as an "eNB-type RSU"; a gNB may be referred to as a "gNB-type RSU"; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

In some embodiments, the RAN **104** may be an LTE RAN **110** with eNBs, for example, eNB **112**. The LTE RAN **110** may provide an LTE air interface with the following characteristics: SCS of 15 kHz; CP-OFDM waveform for DL and SC-FDMA waveform for UL; turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operating on sub-6 GHz bands.

In some embodiments, the RAN **104** may be an NG-RAN **114** with gNBs, for example, gNB **116**, or ng-eNBs, for example, ng-eNB **118**. The gNB **116** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **116** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **118** may also connect with the 5G core through an NG interface, but may connect with a UE via an LTE air interface. The gNB **116** and the ng-eNB **118** may connect with each other over an Xn interface.

In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **114** and a UPF **148** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **114** and an AMF **144** (e.g., N2 interface).

The NG-RAN **114** may provide a 5G-NR air interface with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS, but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH; and tracking reference signal for time tracking. The 5G-NR air interface may operating on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include an SSB that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

In some embodiments, the 5G-NR air interface may utilize BWPs for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the

UE **102** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **102**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **102** with different amount of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with small traffic load while allowing power saving at the UE **102** and in some cases at the gNB **116**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic load.

The RAN **104** is communicatively coupled to CN **120** that includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **102**). The components of the CN **120** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **120** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **120** may be referred to as a network slice, and a logical instantiation of a portion of the CN **120** may be referred to as a network sub-slice.

In some embodiments, the CN **120** may be an LTE CN **122**, which may also be referred to as an EPC. The LTE CN **122** may include MME **124**, SGW **126**, SGSN **128**, HSS **130**, PGW **132**, and PCRF **134** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the LTE CN **122** may be briefly introduced as follows.

The MME **124** may implement mobility management functions to track a current location of the UE **102** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

The SGW **126** may terminate an S1 interface toward the RAN and route data packets between the RAN and the LTE CN **122**. The SGW **126** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

The SGSN **128** may track a location of the UE **102** and perform security functions and access control. In addition, the SGSN **128** may perform inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **124**; MME selection for handovers; etc. The S3 reference point between the MME **124** and the SGSN **128** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

The HSS **130** may include a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **130** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **130** and the MME **124** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **120**.

The PGW **132** may terminate an SGi interface toward a data network (DN) **136** that may include an application/content server **138**. The PGW **132** may route data packets between the LTE CN **122** and the data network **136**. The PGW **132** may be coupled with the SGW **126** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **132** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **132** and the data network **1 36** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. The PGW **132** may be coupled with a PCRF **134** via a Gx reference point.

The PCRF **134** is the policy and charging control element of the LTE CN **122**. The PCRF **134** may be communicatively coupled to the app/content server **138** to determine appropriate QoS and charging parameters for service flows. The PCRF **132** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

In some embodiments, the CN **120** may be a 5GC **140**. The 5GC **140** may include an AUSF **142**, AMF **144**, SMF **146**, UPF **148**, NSSF **150**, NEF **152**, NRF **154**, PCF **156**, UDM **158**, and AF **160** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the 5GC **140** may be briefly introduced as follows.

The AUSF **142** may store data for authentication of UE **102** and handle authentication-related functionality. The AUSF **142** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **140** over reference points as shown, the AUSF **142** may exhibit an Nausf service-based interface.

The AMF **144** may allow other functions of the 5GC **140** to communicate with the UE **102** and the RAN **104** and to subscribe to notifications about mobility events with respect to the UE **102**. The AMF **144** may be responsible for registration management (for example, for registering UE **102**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **144** may provide transport for SM messages between the UE **102** and the SMF **146**, and act as a transparent proxy for routing SM messages. AMF **144** may also provide transport for SMS messages between UE **102** and an SMSF. AMF **144** may interact with the AUSF **142** and the UE **102** to perform various security anchor and context management functions. Furthermore, AMF **144** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **104** and the AMF **144**; and the AMF **144** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **144** may also support NAS signaling with the UE **102** over an N3 IWF interface.

The SMF **146** may be responsible for SM (for example, session establishment, tunnel management between UPF **148** and AN **108**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **148** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **144** over N2 to AN **108**; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **102** and the data network **136**.

The UPF **148** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to data network **136**, and a branching point to support multi-homed PDU session. The UPF **148** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **148** may include an uplink classifier to support routing traffic flows to a data network.

The NSSF **150** may select a set of network slice instances serving the UE **102**. The NSSF **150** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **150** may also determine the AMF set to be used to serve the UE **102**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **154**. The selection of a set of network slice instances for the UE **102** may be triggered by the AMF **144** with which the UE **102** is registered by interacting with the NSSF **150**, which may lead to a change of AMF. The NSSF **150** may interact with the AMF **144** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **150** may exhibit an Nnssf service-based interface.

The NEF **152** may securely expose services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, AFs (e.g., AF **160**), edge computing or fog computing systems, etc. In such embodiments, the NEF **152** may authenticate, authorize, or throttle the AFs. NEF **152** may also translate information exchanged with the AF **160** and information exchanged with internal network functions. For example, the NEF **152** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **152** may also receive information from other NFs based on exposed capabilities of other NFs. This information may be stored at the NEF **152** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **152** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **152** may exhibit an Nnef service-based interface.

The NRF **154** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **154** also maintains information of available NF instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **154** may exhibit the Nnrf service-based interface.

The PCF **156** may provide policy rules to control plane functions to enforce them, and may also support unified policy framework to govern network behavior. The PCF **156** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **158**. In addition to communicating with functions over reference points as shown, the PCF **156** exhibit an Npcf service-based interface.

The UDM **158** may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE **102**. For example, subscription data may be communicated via an N8 reference point between the UDM **158** and the AMF **144**. The UDM **158** may include two parts, an application front end and a UDR. The UDR may store subscription data and policy data for the UDM **158** and the PCF **156**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **102**) for the NEF **152**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **158**, PCF **156**, and NEF **152** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **158** may exhibit the Nudm service-based interface.

The AF **160** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

In some embodiments, the 5GC **140** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **102** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **140** may select a UPF **148** close to the UE **102** and execute traffic steering from the UPF **148** to data network **136** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **160**. In this way, the AF **160** may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF **160** is considered to be a trusted entity, the network operator may permit AF **160** to interact directly with relevant NFs. Additionally, the AF **160** may exhibit an Naf service-based interface.

The data network **136** may represent various network operator services, Internet access, or third party services that may be provided by one or more servers including, for example, application/content server **138**.

FIG. **2** schematically illustrates a wireless network **200** in accordance with various embodiments. The wireless network **200** may include a UE **202** in wireless communication with an AN **204**. The UE **202** and AN **204** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

The UE **202** may be communicatively coupled with the AN **204** via connection **206**. The connection **206** is illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHz frequencies.

The UE **202** may include a host platform **208** coupled with a modem platform **210**. The host platform **208** may include application processing circuitry **212**, which may be coupled with protocol processing circuitry **214** of the modem platform **210**. The application processing circuitry **212** may run various applications for the UE **202** that source/sink application data. The application processing circuitry **212** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example, IP) operations

The protocol processing circuitry **214** may implement one or more of layer operations to facilitate transmission or reception of data over the connection **206**. The layer operations implemented by the protocol processing circuitry **214** may include, for example, MAC, RLC, PDCP, RRC and NAS operations.

The modem platform **210** may further include digital baseband circuitry **216** that may implement one or more layer operations that are "below" layer operations performed by the protocol processing circuitry **214** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

The modem platform **210** may further include transmit circuitry **218**, receive circuitry **220**, RF circuitry **222**, and RF front end (RFFE) **224**, which may include or connect to one or more antenna panels **226**. Briefly, the transmit circuitry **218** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **220** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **222** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **224** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **218**, receive circuitry **220**, RF circuitry **222**, RFFE **224**, and antenna panels **226** (referred generically as "transmit/receive components") may be specific to details of a specific implementation such as, for example, whether communication is TDM or FDM, in mmWave or sub-6 gHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed in the same or different chips/modules, etc.

In some embodiments, the protocol processing circuitry **214** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

A UE reception may be established by and via the antenna panels **226**, RFFE **224**, RF circuitry **222**, receive circuitry **220**, digital baseband circuitry **216**, and protocol processing circuitry **214**. In some embodiments, the antenna panels **226** may receive a transmission from the AN **204** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **226**.

A UE transmission may be established by and via the protocol processing circuitry **214**, digital baseband circuitry **216**, transmit circuitry **218**, RF circuitry **222**, RFFE **224**, and antenna panels **226**. In some embodiments, the transmit components of the UE **204** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **226**.

Similar to the UE **202**, the AN **204** may include a host platform **228** coupled with a modem platform **230**. The host platform **228** may include application processing circuitry **232** coupled with protocol processing circuitry **234** of the modem platform **230**. The modem platform may further include digital baseband circuitry **236**, transmit circuitry **238**, receive circuitry **240**, RF circuitry **242**, RFFE circuitry **244**, and antenna panels **246**. The components of the AN **204** may be similar to and substantially interchangeable with like-named components of the UE **202**. In addition to performing data transmission/reception as described above, the components of the AN **208** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

FIG. **3** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **3** shows a diagrammatic representation of hardware resources **300** including one or more processors (or processor cores) **310**, one or more memory/storage devices **320**, and one or more communication resources **330**, each of which may be communicatively coupled via a bus **340** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **302** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **300**.

The processors **310** may include, for example, a processor **312** and a processor **314**. The processors **310** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices **320** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **320** may include, but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources **330** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **304** or one or more databases **306** or other network elements via a network **308**. For example, the communication resources **330** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

Instructions **350** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **310** to perform any one or more of the methodologies discussed herein. The instructions **350** may reside, completely or partially, within at least one of the processors **310** (e.g., within the processor's cache memory), the memory/storage devices **320**, or any suitable combination thereof. Furthermore, any portion of the instructions **350** may be transferred to the hardware resources **300** from any combination of the peripheral devices **304** or the databases **306**. Accordingly, the memory of processors **310**, the memory/storage devices **320**, the peripheral devices **304**, and the databases **306** are examples of computer-readable and machine-readable media.

Example Procedures

In some embodiments, the electronic device(s), network(s), system(s), chip(s) or component(s), or portions or implementations thereof, of FIGS. **1-3**, or some other figure herein, may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof.

For example, FIG. **4** illustrates an example process **400** in accordance with various embodiments. The process **400** may be performed by a UE or a portion thereof. At **402**, the process **400** may include identifying a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i},$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i. At **404**, the process may further include performing one or more PRS measurements within the PRS measurement period.

FIG. **5** illustrates another example process **500** in accordance with various embodiments. The process **500** may be performed by a gNB or a portion thereof. At **502**, the process **500** may include encoding, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i},$$

wherein $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is the periodicity of the downlink PRS resource on the positioning frequency layer i. At **504**, the process **500** may further include receiving, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

Examples

Example A1 may include one or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a user equipment (UE) configure the UE to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i},$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i; and perform one or more PRS measurements within the PRS measurement period.

Example A2 may include the one or more NTCRM of example A1, wherein the PRS measurement period is for a single positioning frequency layer.

Example A3 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

Example A4 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

Example A5 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to decode a broadcast message to determine a value of $T_{PRS,i}$.

Example A6 may include the one or more NTCRM of example A1, wherein the one or more PRS measurements are performed without a measurement gap.

Example A7 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.

Example A8 may include one or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a next generation Node B (gNB) configure the gNB to: encode, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i},$$

wherein $T_{effect,i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is the periodicity of the downlink PRS resource on the positioning frequency layer i; and receive, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

Example A9 may include the one or more NTCRM of example A8, wherein the PRS measurement period is for a single positioning frequency layer.

Example A10 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

Example A11 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement.

Example A12 may include the one or more NTCRM of example A8, wherein the indication of the periodicity of the downlink PRS message is transmitted via broadcast message to a plurality of UEs.

Example A13 may include the one or more NTCRM of example A8, wherein the one or more PRS measurements are performed without a measurement gap.

Example A14 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to: encode, for transmission to the UE, a message that activates a measurement gap for the one or more PRS measurements, wherein the UE is to use the measurement gap upon expiration of an activation delay time.

Example A15 may include an apparatus to be implemented in a user equipment (UE), the apparatus comprising: a memory to store a value, $T_{PRS,i}$, of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i; and processor circuitry coupled to the memory. The processor circuitry is to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i},$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is an effective periodicity of a PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{PRS,i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i; and perform one or more PRS measurements within the PRS measurement period.

Example A16 may include the apparatus of example A15, wherein the PRS measurement period is for a single positioning frequency layer.

Example A17 may include the apparatus of example A15, wherein the processor circuitry is further to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

Example A18 may include the apparatus of example A15, wherein the processor circuitry is further to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

Example A19 may include the apparatus of example A15, wherein the one or more PRS measurements are performed without a measurement gap.

Example A20 may include the apparatus of example A15, wherein the processor circuitry is further to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.

Example B1 may include a method to define the requirements for UE positioning measurements without gap.

Example B2 may include the method of example B1 or some other example herein, wherein the basic timing interval to be used to define PRS measurement period can be:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i}$$

Example B3 may include the method of example B1 or some other example herein, wherein the requirements are applicable for the single positioning frequency layer (PFL) measurements.

Example B4 may include the method of example B1 or some other example herein, where when multiple PFLs configured, UE can decide perform the intra-f measurement for the PFL which are contained within UE's active BWP. And other PFLs by inter-f measurement.

Example B5 may include the method to define the requirements for UE positioning measurements with pre-configured gap.

Example B6 may include the method of example B5 or some other example herein, wherein the requirements on pre-configured MG based measurement can include the pre-configured MG activation delay

Example B7 may include the method of example B5 or some other example herein, wherein the requirements on pre-configured MG based measurement can include the pre-configured MG activation/deactivation transition time.

Example B8 may include a method of a UE, the method comprising:

determining a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i};$$

performing one or more PRS measurements in the PRS measurement period.

Example B9 may include the method of example B8 or some other example herein, wherein the PRS measurement period is for a single positioning frequency layer (PFL).

Example B10 may include the method of example B8-B9 or some other example herein, further comprising determining to perform one or more intra-frequency measurements for a PFL which included in an active BWP of the UE, and/or performing PRS measurements on one or more other. PFLs by inter-frequency measurement.

Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.

Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.

Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.

Example Z04 may include a method, technique, or process as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof.

Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.

Example Z06 may include a signal as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof.

Example Z07 may include a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z08 may include a signal encoded with data as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z09 may include a signal encoded with a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z10 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.

Example Z11 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.

Example Z12 may include a signal in a wireless network as shown and described herein.

Example Z13 may include a method of communicating in a wireless network as shown and described herein.

Example Z14 may include a system for providing wireless communication as shown and described herein.

Example Z15 may include a device for providing wireless communication as shown and described herein.

Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

ABBREVIATIONS

Unless used differently herein, terms, definitions, and abbreviations may be consistent with terms, definitions, and abbreviations defined in 3GPP TR 21.905 v16.0.0 (2019-06). For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein.

3GPP	Third Generation Partnership Project
4G	Fourth Generation
5G	Fifth Generation
5GC	5G Core network
AC	Application Client
ACR	Application Context Relocation
ACK	Acknowledgement
ACID	Application Client Identification
AF	Application Function
AM	Acknowledged Mode
AMBR	Aggregate Maximum Bit Rate
AMF	Access and Mobility Management Function
AN	Access Network
ANR	Automatic Neighbour Relation
AOA	Angle of Arrival
AP	Application Protocol, Antenna Port, Access Point
API	Application Programming Interface
APN	Access Point Name
ARP	Allocation and Retention Priority
ARQ	Automatic Repeat Request
AS	Access Stratum
ASP	Application Service Provider
ASN.1	Abstract Syntax Notation One
AUSF	Authentication Server Function
AWGN	Additive White Gaussian Noise
BAP	Backhaul Adaptation Protocol
BCH	Broadcast Channel
BER	Bit Error Ratio
BFD	Beam Failure Detection
BLER	Block Error Rate
BPSK	Binary Phase Shift Keying
BRAS	Broadband Remote Access Server
BSS	Business Support System
BS	Base Station
BSR	Buffer Status Report
BW	Bandwidth
BWP	Bandwidth Part
C-RNTI	Cell Radio Network Temporary Identity
CA	Carrier Aggregation, Certification Authority
CAPEX	CAPital EXpenditure
CBRA	Contention Based Random Access
CC	Component Carrier, Country Code, Cryptographic Checksum
CCA	Clear Channel Assessment
CCE	Control Channel Element

-continued

CCCH	Common Control Channel
CE	Coverage Enhancement
CDM	Content Delivery Network
CDMA	Code-Division Multiple Access
CDR	Charging Data Request
CDR	Charging Data Response
CFRA	Contention Free Random Access
CG	Cell Group
CGF	Charging Gateway Function
CHF	Charging Function
CI	Cell Identity
CID	Cell-ID (e.g., positioning method)
CIM	Common Information Model
CIR	Carrier to Interference Ratio
CK	Cipher Key
CM	Connection Management, Conditional Mandatory
CMAS	Commercial Mobile Alert Service
CMD	Command
CMS	Cloud Management System
CO	Conditional Optional
CoMP	Coordinated Multi-Point
CORESET	Control Resource Set
COTS	Commercial Off-The-Shelf
CP	Control Plane, Cyclic Prefix, Connection Point
CPD	Connection Point Descriptor
CPE	Customer Premise Equipment
CPICH	Common Pilot Channel
CQI	Channel Quality Indicator
CPU	CSI processing unit, Central Processing Unit
C/R	Command/Response field bit
CRAN	Cloud Radio Access Network, Cloud RAN
CRB	Common Resource Block
CRC	Cyclic Redundancy Check
CRI	Channel-State Information Resource Indicator, CSI-RS Resource Indicator
C-RNTI	Cell RNTI
CS	Circuit Switched
CSCF	call session control function
CSAR	Cloud Service Archive
CSI	Channel-State Information

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CSI-IM	CSI Interference Measurement
CSI-RS	CSI Reference Signal
CSI-RSRP	CSI reference signal received power
CSI-RSRQ	CSI reference signal received quality
CSI-SINR	CSI signal-to-noise and interference ratio
CSMA	Carrier Sense Multiple Access
CSMA/CA	CSMA with collision avoidance
CSS	Common Search Space, Cell-specific Search Space
CTF	Charging Trigger Function
CTS	Clear-to-Send
CW	Codeword
CWS	Contention Window Size
D2D	Device-to-Device
DC	Dual Connectivity, Direct Current
DCI	Downlink Control Information
DF	Deployment Flavour
DL	Downlink
DMTF	Distributed Management Task Force
DPDK	Data Plane Development Kit
DM-RS, DMRS	Demodulation Reference Signal
DN	Data network
DNN	Data Network Name
DNAI	Data Network Access Identifier
DRB	Data Radio Bearer
DRS	Discovery Reference Signal
DRX	Discontinuous Reception
DSL	Domain Specific Language. Digital Subscriber Line
DSLAM	DSL Access Multiplexer
DwPTS	Downlink Pilot Time Slot
E-LAN	Ethernet Local Area Network
E2E	End-to-End
EAS	Edge Application Server
ECCA	extended clear channel assessment, extended CCA
ECCE	Enhanced Control Channel Element, Enhanced CCE
ED	Energy Detection
EDGE	Enhanced Datarates for GSM Evolution (GSM Evolution)
EAS	Edge Application Server
EASID	Edge Application Server Identification
ECS	Edge Configuration Server

-continued

ECSP	Edge Computing Service Provider
EDN	Edge Data Network
EEC	Edge Enabler Client
EECID	Edge Enabler Client Identification
EES	Edge Enabler Server
EESID	Edge Enabler Server Identification
EHE	Edge Hosting Environment
EGMF	Exposure Governance Management Function
EGPRS	Enhanced GPRS
EIR	Equipment Identity Register
eLAA	enhanced Licensed Assisted Access, enhanced LAA
EM	Element Manager
eMBB	Enhanced Mobile Broadband
EMS	Element Management System
eNB	evolved NodeB, E-UTRAN Node B
EN-DC	E-UTRA-NR Dual Connectivity
EPC	Evolved Packet Core
EPDCCH	enhanced PDCCH, enhanced Physical Downlink Control Cannel
EPRE	Energy per resource element
EPS	Evolved Packet System
EREG	enhanced REG, enhanced resource element groups
ETSI	European Telecommunications Standards Institute
ETWS	Earthquake and Tsunami Warning System
eUICC	embedded UICC, embedded Universal Integrated Circuit Card
E-UTRA	Evolved UTRA
E-UTRAN	Evolved UTRAN
EV2X	Enhanced V2X
F1AP	F1 Application Protocol
F1-C	F1 Control plane interface
F1-U	F1 User plane interface
FACCH	Fast Associated Control CHannel
FACCH/F	Fast Associated Control Channel/Full rate
FACCH/H	Fast Associated Control Channel/Half rate

-continued

FACH	Forward Access Channel
FAUSCH	Fast Uplink Signalling Channel
FB	Functional Block
FBI	Feedback Information
FCC	Federal Communications Commission
FCCH	Frequency Correction CHannel
FDD	Frequency Division Duplex
FDM	Frequency Division Multiplex
FDMA	Frequency Division Multiple Access
FE	Front End
FEC	Forward Error Correction
FFS	For Further Study
FFT	Fast Fourier Transformation
feLAA	further enhanced Licensed Assisted Access, further enhanced LAA
FN	Frame Number
FPGA	Field-Programmable Gate Array
FR	Frequency Range
FQDN	Fully Qualified Domain Name
G-RNTI	GERAN Radio Network Temporary Identity
GERAN	GSM EDGE RAN, GSM EDGE Radio Access Network
GGSN	Gateway GPRS Support Node
GLONASS	GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (Engl.: Global Navigation Satellite System)
gNB	Next Generation NodeB
gNB-CU	gNB-centralized unit, Next Generation NodeB centralized unit
gNB-DU	gNB-distributed unit, Next Generation NodeB distributed unit
GNSS	Global Navigation Satellite System
GPRS	General Packet Radio Service
GPSI	Generic Public Subscription Identifier
GSM	Global System for Mobile Communications, Groupe Spécial Mobile
GTP	GPRS Tunneling Protocol
GTP-UGPRS	Tunnelling Protocol for User Plane
GTS	Go To Sleep Signal (related to WUS)
GUMMEI	Globally Unique MME Identifier

-continued

GUTI	Globally Unique Temporary UE Identity
HARQ	Hybrid ARQ, Hybrid Automatic Repeat Request
HANDO	Handover
HFN	HyperFrame Number
HHO	Hard Handover
HLR	Home Location Register
HN	Home Network
HO	Handover
HPLMN	Home Public Land Mobile Network
HSDPA	High Speed Downlink Packet Access
HSN	Hopping Sequence Number
HSPA	High Speed Packet Access
HSS	Home Subscriber Server
HSUPA	High Speed Uplink Packet Access
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure (https is http/1.1 over SSL, i.e. port 443)
I-Block	Information Block
ICCID	Integrated Circuit Card Identification
IAB	Integrated Access and Backhaul
ICIC	Inter-Cell Interference Coordination
ID	Identity, identifier
IDFT	Inverse Discrete Fourier Transform
IE	Information element
IBE	In-Band Emission
IEEE	Institute of Electrical and Electronics Engineers
IEI	Information Element Identifier
IEIDL	Information Element Identifier Data Length
IETF	Internet Engineering Task Force
IF	Infrastructure
IIOT	Industrial Internet of Things
IM	Interference Measurement, Intermodulation, IP Multimedia
IMC	IMS Credentials
IMEI	International Mobile Equipment Identity
IMGI	International mobile group identity
IMPI	IP Multimedia Private Identity
IMPU	IP Multimedia PUblic identity
IMS	IP Multimedia Subsystem
IMSI	International Mobile Subscriber Identity

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IoT	Internet of Things
IP	Internet Protocol
Ipsec	IP Security, Internet Protocol Security
IP-CAN	IP-Connectivity Access Network
IP-M	IP Multicast
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
IR	Infrared
IS	In Sync
IRP	Integration Reference Point
ISDN	Integrated Services Digital Network
ISIM	IM Services Identity Module
ISO	International Organisation for Standardisation
ISP	Internet Service Provider
IWF	Interworking-Function
I-WLAN	Interworking WLAN
	Constraint length of the convolutional code, USIM Individual key
kB	Kilobyte (1000 bytes)
kbps	kilo-bits per second
Kc	Ciphering key
Ki	Individual subscriber authentication key
KPI	Key Performance Indicator
KQI	Key Quality Indicator
KSI	Key Set Identifier
ksps	kilo-symbols per second
KVM	Kernel Virtual Machine
L1	Layer 1 (physical layer)
L1-RSRP	Layer 1 reference signal received power
L2	Layer 2 (data link layer)
L3	Layer 3 (network layer)
LAA	Licensed Assisted Access
LAN	Local Area Network
LADN	Local Area Data Network
LBT	Listen Before Talk
LCM	LifeCycle Management
LCR	Low Chip Rate
LCS	Location Services
LCID	Logical Channel ID
LI	Layer Indicator
LLC	Logical Link Control, Low Layer Compatibility
LMF	Location Management Function
LOS	Line of Sight
LPLMN	Local PLMN
LPP	LTE Positioning Protocol

LSB	Least Significant Bit
LTE	Long Term Evolution
LWA	LTE-WLAN aggregation
LWIP	LTE/WLAN Radio Level Integration with IPsec Tunnel
LTE	Long Term Evolution
M2M	Machine-to-Machine
MAC	Medium Access Control (protocol layering context)
MAC	Message authentication code (security/encryption context)
MAC-A	MAC used for authentication and key agreement (TSG T WG3 context)
MAC-IMAC	used for data integrity of signalling messages (TSG T WG3 context)
MANO	Management and Orchestration
MBMS	Multimedia Broadcast and Multicast Service
MBSFN	Multimedia Broadcast multicast service Single Frequency Network
MCC	Mobile Country Code
MCG	Master Cell Group
MCOT	Maximum Channel Occupancy Time
MCS	Modulation and coding scheme
MDAF	Management Data Analytics Function
MDAS	Management Data Analytics Service
MDT	Minimization of Drive Tests
ME	Mobile Equipment
MeNB	master eNB
MER	Message Error Ratio
MGL	Measurement Gap Length
MGRP	Measurement Gap Repetition Period
MIB	Master Information Block, Management Information Base
MIMO	Multiple Input Multiple Output
MLC	Mobile Location Centre
MM	Mobility Management
MME	Mobility Management Entity
MN	Master Node
MNO	Mobile Network Operator
MO	Measurement Object, Mobile Originated
MPBCH	MTC Physical Broadcast CHannel
MPDCCH	MTC Physical Downlink Control CHannel

MPDSCH	MTC Physical Downlink Shared CHannel
MPRACH	MTC Physical Random Access CHannel
MPUSCH	MTC Physical Uplink Shared Channel
MPLS	MultiProtocol Label Switching
MS	Mobile Station
MSB	Most Significant Bit
MSC	Mobile Switching Centre
MSI	Minimum System Information, MCH Scheduling Information
MSID	Mobile Station Identifier
MSIN	Mobile Station Identification Number
MSISDN	Mobile Subscriber ISDN Number
MT	Mobile Terminated, Mobile Termination
MTC	Machine-Type Communications
mMTC	massive MTC, massive Machine-Type Communications
MU-MIMO	Multi User MIMO
MWUS	MTC wake-up signal, MTC WUS
NACK	Negative Acknowledgement
NAI	Network Access Identifier
NAS	Non-Access Stratum, Non-Access Stratum layer
NCT	Network Connectivity Topology
NC-JT	Non-coherent Joint Transmission
NEC	Network Capability Exposure
NE-DC	NR-E-UTRA Dual Connectivity
NEF	Network Exposure Function
NF	Network Function
NFP	Network Forwarding Path
NFPD	Network Forwarding Path Descriptor
NFV	Network Functions Virtualization
NFVI	NFV Infrastructure
NFVO	NFV Orchestrator
NG	Next Generation, Next Gen
NGEN-DC	NG-RAN E-UTRA-NR Dual Connectivity
NM	Network Manager
NMS	Network Management System
N-PoP	Network Point of Presence
NMIB,	N-MIB Narrowband MIB

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NPBCH	Narrowband Physical Broadcast CHannel
NPDCCH	Narrowband Physical Downlink Control CHannel
NPDSCH	Narrowband Physical Downlink Shared CHannel
NPRACH	Narrowband Physical Random Access CHannel
NPUSCH	Narrowband Physical Uplink Shared CHannel
NPSS	Narrowband Primary Synchronization Signal
NSSS	Narrowband Secondary Synchronization Signal
NR	New Radio, Neighbour Relation
NRF	NF Repository Function
NRS	Narrowband Reference Signal
NS	Network Service
NSA	Non-Standalone operation mode
NSD	Network Service Descriptor
NSR	Network Service Record
NSSAI	Network Slice Selection Assistance Information
S-NNSAI	Single-NSSAI
NSSF	Network Slice Selection Function
NW	Network
NWUS	Narrowband wake-up signal, Narrowband WUS
NZP	Non-Zero Power
O&M	Operation and Maintenance
ODU2	Optical channel Data Unit-type 2
OFDM	Orthogonal Frequency Division Multiplexing
OFDMA	Orthogonal Frequency Division Multiple Access
OOB	Out-of-band
OOS	Out of Sync
OPEX	OPerating EXpense
OSI	Other System Information
OSS	Operations Support System
OTA	over-the-air
PAPR	Peak-to-Average Power Ratio
PAR	Peak to Average Ratio
PBCH	Physical Broadcast Channel
PC	Power Control, Personal Computer
PCC	Primary Component Carrier, Primary CC

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P-CSCF	Proxy CSCF
PCell	Primary Cell
PCI	Physical Cell ID, Physical Cell Identity
PCEF	Policy and Charging Enforcement Function
PCF	Policy Control Function
PCRF	Policy Control and Charging Rules Function
PDCP	Packet Data Convergence Protocol, Packet Data Convergence Protocol layer
PDCCH	Physical Downlink Control Channel
PDCP	Packet Data Convergence Protocol
PDN	Packet Data Network, Public Data Network
PDSCH	Physical Downlink Shared Channel
PDU	Protocol Data Unit
PEI	Permanent Equipment Identifiers
PFD	Packet Flow Description
P-GW	PDN Gateway
PHICH	Physical hybrid-ARQ indicator channel
PHY	Physical layer
PLMN	Public Land Mobile Network
PIN	Personal Identification Number
PM	Performance Measurement
PMI	Precoding Matrix Indicator
PNF	Physical Network Function
PNFD	Physical Network Function Descriptor
PNFR	Physical Network Function Record
POC	PTT over Cellular PP, PTP Point-to-Point
PPP	Point-to-Point Protocol
PRACH	Physical RACH
PRB	Physical resource block
PRG	Physical resource block group
ProSe	Proximity Services, Proximity-Based Service
PRS	Positioning Reference Signal
PRR	Packet Reception Radio
PS	Packet Services
PSBCH	Physical Sidelink Broadcast Channel
PSDCH	Physical Sidelink Downlink Channel
PSCCH	Physical Sidelink Control Channel

-continued

PSSCH	Physical Sidelink Shared Channel
PSCell	Primary SCell
PSS	Primary Synchronization Signal
PSTN	Public Switched Telephone Network
PT-RS	Phase-tracking reference signal
PTT	Push-to-Talk
PUCCH	Physical Uplink Control Channel
PUSCH	Physical Uplink Shared Channel
QAM	Quadrature Amplitude Modulation
QCI	QoS class of identifier
QCL	Quasi co-location
QFI	QoS Flow ID, QoS Flow Identifier
QoS	Quality of Service
QPSK	Quadrature (Quaternary) Phase Shift Keying
QZSS	Quasi-Zenith Satellite System
RA-RNTI	Random Access RNTI
RAB	Radio Access Bearer, Random Access Burst
RACH	Random Access Channel
RADIUS	Remote Authentication Dial In User Service
RAN	Radio Access Network
RAND	RANDom number (used for authentication)
RAR	Random Access Response
RAT	Radio Access Technology
RAU	Routing Area Update
RB	Resource block, Radio Bearer
RBG	Resource block group
REG	Resource Element Group
Rel	Release
REQ	REQuest
RF	Radio Frequency
RI	Rank Indicator
RIV	Resource indicator value
RL	Radio Link
RLC	Radio Link Control, Radio Link Control layer
RLC AM	RLC Acknowledged Mode
RLC UM	RLC Unacknowledged Mode
RLF	Radio Link Failure
RLM	Radio Link Monitoring
RLM-RS	Reference Signal for RLM
RM	Registration Management

-continued

RMC	Reference Measurement Channel
RMSI	Remaining MSI, Remaining Minimum System Information
RN	Relay Node
RNC	Radio Network Controller
RNL	Radio Network Layer
RNTI	Radio Network Temporary Identifier
ROHC	RObust Header Compression
RRC	Radio Resource Control, Radio Resource Control layer
RRM	Radio Resource Management
RS	Reference Signal
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
RSU	Road Side Unit
RSTD	Reference Signal Time difference
RTP	Real Time Protocol
RTS	Ready-To-Send
RTT	Round Trip Time
Rx	Reception, Receiving, Receiver
S1AP	S1 Application Protocol
S1-MME	S1 for the control plane
S1-U	S1 for the user plane
S-CSCF	serving CSCF
S-GW	Serving Gateway
S-RNTI	SRNC Radio Network Temporary Identity
S-TMSI	SAE Temporary Mobile Station Identifier
SA	Standalone operation mode
SAE	System Architecture Evolution
SAP	Service Access Point
SAPD	Service Access Point Descriptor
SAPI	Service Access Point Identifier
SCC	Secondary Component Carrier, Secondary CC
SCell	Secondary Cell
SCEF	Service Capability Exposure Function
SC-FDMA	Single Carrier Frequency Division Multiple Access
SCG	Secondary Cell Group
SCM	Security Context Management
SCS	Subcarrier Spacing
SCTP	Stream Control Transmission Protocol

-continued

SDAP	Service Data Adaptation Protocol, Service Data Adaptation Protocol layer
SDL	Supplementary Downlink
SDNF	Structured Data Storage Network Function
SDP	Session Description Protocol
SDSF	Structured Data Storage Function
SDT	Small Data Transmission
SDU	Service Data Unit
SEAF	Security Anchor Function
SeNB	secondary eNB
SEPP	Security Edge Protection Proxy
SFI	Slot format indication
SFTD	Space-Frequency Time Diversity, SFN and frame timing difference
SFN	System Frame Number
SgNB	Secondary gNB
SGSN	Serving GPRS Support Node
S-GW	Serving Gateway
SI	System Information
SI-RNTI	System Information RNTI
SIB	System Information Block
SIM	Subscriber Identity Module
SIP	Session Initiated Protocol
SiP	System in Package
SL	Sidelink
SLA	Service Level Agreement
SM	Session Management
SMF	Session Management Function
SMS	Short Message Service
SMSF	SMS Function
SMTC	SSB-based Measurement Timing Configuration
SN	Secondary Node, Sequence Number
SoC	System on Chip
SON	Self-Organizing Network
SpCell	Special Cell
SP-CSI-RNTI	Semi-Persistent CSI RNTI
SPS	Semi-Persistent Scheduling
SQN	Sequence number
SR	Scheduling Request
SRB	Signalling Radio Bearer
SRS	Sounding Reference Signal
SS	Synchronization Signal
SSB	Synchronization Signal Block
SSID	Service Set Identifier

-continued

SS/PBCH	Block
SSBRI	SS/PBCH Block Resource Indicator, Synchronization Signal Block Resource Indicator
SSC	Session and Service Continuity
SS-RSRP	Synchronization Signal based Reference Signal Received Power
SS-RSRQ	Synchronization Signal based Reference Signal Received Quality
SS-SINR	Synchronization Signal based Signal to Noise and Interference Ratio
SSS	Secondary Synchronization Signal
SSSG	Search Space Set Group
SSSIF	Search Space Set Indicator
SST	Slice/Service Types
SU-MIMO	Single User MIMO
SUL	Supplementary Uplink
TA	Timing Advance, Tracking Area
TAC	Tracking Area Code
TAG	Timing Advance Group
TAI	Tracking Area Identity
TAU	Tracking Area Update
TB	Transport Block
TBS	Transport Block Size
TBD	To Be Defined
TCI	Transmission Configuration Indicator
TCP	Transmission Communication Protocol
TDD	Time Division Duplex
TDM	Time Division Multiplexing
TDMA	Time Division Multiple Access
TE	Terminal Equipment
TEID	Tunnel End Point Identifier
TFT	Traffic Flow Template
TMSI	Temporary Mobile Subscriber Identity
TNL	Transport Network Layer
TPC	Transmit Power Control
TPMI	Transmitted Precoding Matrix Indicator
TR	Technical Report
TRP, TRxP	Transmission Reception Point
TRS	Tracking Reference Signal
TRx	Transceiver
TS	Technical Specifications, Technical Standard

-continued

TTI	Transmission Time Interval
Tx	Transmission, Transmitting, Transmitter
U-RNTI	UTRAN Radio Network Temporary Identity
UART	Universal Asynchronous Receiver and Transmitter
UCI	Uplink Control Information
UE	User Equipment
UDM	Unified Data Management
UDP	User Datagram Protocol
UDSF	Unstructured Data Storage Network Function
UICC	Universal Integrated Circuit Card
UL	Uplink
UM	Unacknowledged Mode
UML	Unified Modelling Language
UMTS	Universal Mobile Telecommunications System
UP	User Plane
UPF	User Plane Function
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URLLC	Ultra-Reliable and Low Latency
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
USS	UE-specific search space
UTRA	UMTS Terrestrial Radio Access
UTRAN	Universal Terrestrial Radio Access Network
UwPTS	Uplink Pilot Time Slot
V2I	Vehicle-to-Infrastruction
V2P	Vehicle-to-Pedestrian
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-everything
VIM	Virtualized Infrastructure Manager
VL	Virtual Link,
VLAN	Virtual LAN, Virtual Local Area Network
VM	Virtual Machine
VNF	Virtualized Network Function
VNFFG	VNF Forwarding Graph
VNFFGD	VNF Forwarding Graph Descriptor
VNFM	VNF Manager
VoIP	Voice-over-IP, Voice-over-Internet Protocol

-continued

VPLMN	Visited Public Land Mobile Network
VPN	Virtual Private Network Network
VRB	Virtual Resource Block
WiMAX	Worldwide Interoperability for Microwave Access
WLAN	Wireless Local Area Network
WMAN	Wireless Metropolitan Area Network
WPAN	Wireless Personal Area Network
X2-C	X2-Control plane
X2-U	X2-User plane
XML	extensible Markup Language
XRES	EXpected user RESponse
XOR	exclusive OR
ZC	Zadoff-Chu
ZP	Zero Power

Terminology

For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein.

The term "circuitry" as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term "circuitry" may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

The term "processor circuitry" as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. Processing circuitry may include one or more processing cores to execute instructions and one or more memory structures to store program and data information. The term "processor circuitry" may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. Processing circuitry may include more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. The terms "application circuitry" and/or "baseband circuitry" may be considered synonymous to, and may be referred to as, "processor circuitry."

The term "interface circuitry" as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term "interface circuitry" may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

The term "user equipment" or "UE" as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term "user equipment" or "UE" may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term "user equipment" or "UE" may include any type of wireless/wired device or any computing device including a wireless communications interface.

The term "network element" as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term "network element" may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

The term "computer system" as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term "computer system" and/or "system" may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term "computer system" and/or "system" may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

The term "appliance," "computer appliance," or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A "virtual appliance" is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

The term "resource" as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A "hardware resource" may refer to compute, storage, and/or network resources provided by physical hardware element(s). A "virtualized resource" may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term "network resource" or "communication resource" may refer to resources that are accessible by computer devices/systems via a communications network. The term "system resources" may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

The term "channel" as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term "channel" may be synonymous with and/or equivalent to "communications channel," "data communications channel," "transmission channel," "data transmission channel," "access channel," "data access channel," "link," "data link," "carrier," "radiofrequency carrier," and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term "link" as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

The terms "instantiate," "instantiation," and the like as used herein refers to the creation of an instance. An "instance" also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

The terms "coupled," "communicatively coupled," along with derivatives thereof are used herein. The term "coupled" may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term "directly coupled" may mean that two or more elements are in direct contact with one another. The term "communicatively coupled" may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or link, and/or the like.

The term "information element" refers to a structural element containing one or more fields. The term "field" refers to individual contents of an information element, or a data element that contains content.

The term "SMTC" refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

The term "SSB" refers to an SS/PBCH block.

The term a "Primary Cell" refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

The term "Primary SCG Cell" refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

The term "Secondary Cell" refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

The term "Secondary Cell Group" refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

The term "Serving Cell" refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

The term “serving cell” or “serving cells” refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/.

The term “Special Cell” refers to the PCell of the MCG or the PSCell of the SCG for DC operation; otherwise, the term “Special Cell” refers to the Pcell.

The invention claimed is:

1. One or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a user equipment (UE) configure the UE to:

identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available\_PRS,i}} \right\rceil * T_{available\_PRS,i}$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, $T_{available\_PRS,i}$ is a least common multiple between $T_{PRS,i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i, and $T_{PRS,i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i; and

perform one or more PRS measurements within the PRS measurement period.

2. The one or more NTCRM of claim **1**, wherein the PRS measurement period is for a single positioning frequency layer.

3. The one or more NTCRM of claim **1**, wherein the instructions, when executed, are further to configure the UE to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

4. The one or more NTCRM of claim **1**, wherein the instructions, when executed, are further to configure the UE to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

5. The one or more NTCRM of claim **1**, wherein the instructions, when executed, are further to configure the UE to decode a broadcast message to determine a value of $T_{PRS,i}$.

6. The one or more NTCRM of claim **1**, wherein the one or more PRS measurements are performed without a measurement gap.

7. The one or more NTCRM of claim **1**, wherein the instructions, when executed, are further to configure the UE to:

decode a message that activates a measurement gap for the one or more PRS measurements; and

use the measurement gap upon expiration of an activation delay time.

8. One or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a next generation Node B (gNB) configure the gNB to:

encode, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource with muting on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{available\_PRS,i}} \right\rceil * T_{available\_PRS,i}$$

wherein $T_{effect,i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, $T_{available\_PRS,i}$ is a least common multiple between $T_{PRS,i}$ and a repetition periodicity of a PRS processing window for measurement in positioning is a least common multiple between $T_{PRS,i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i, and $T_{PRS,i}$ the periodicity of the downlink PRS resource with muting on the positioning frequency layer i; and

receive, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

9. The one or more NTCRM of claim **8**, wherein the PRS measurement period is for a single positioning frequency layer.

10. The one or more NTCRM of claim **8**, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

11. The one or more NTCRM of claim **8**, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement.

12. The one or more NTCRM of claim **8**, wherein the indication of the periodicity of the downlink PRS message is transmitted via broadcast message to a plurality of UEs.

13. The one or more NTCRM of claim **8**, wherein the one or more PRS measurements are performed without a measurement gap.

14. The one or more NTCRM of claim **8**, wherein the instructions, when executed, are further to configure the gNB to:

encode, for transmission to the UE, a message that activates a measurement gap for the one or more PRS measurements, wherein the UE is to use the measurement gap upon expiration of an activation delay time.

15. An apparatus to be implemented in a user equipment (UE), the apparatus comprising:

a memory to store a value, $T_{PRS,i}$, of a periodicity of a downlink positioning reference signal (PRS) resource with muting on a positioning frequency layer i; and

processor circuitry coupled to the memory, the processor circuitry to:

identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{effect,i} = \left\lceil \frac{T_i}{T_{PRS,i}} \right\rceil * T_{PRS,i}$$

wherein i is a positioning frequency layer index, $T_{effect,i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i, T_i corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{available\ PRS,i}$ is a least common multiple between $T_{PRS,i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i, and perform one or more PRS measurements in the PRS measurement period.

16. The apparatus of claim **15**, wherein the PRS measurement period is for a single positioning frequency layer.

17. The apparatus of claim **15**, wherein the processor circuitry is further to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

18. The apparatus of claim **15**, wherein the processor circuitry is further to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

19. The apparatus of claim **15**, wherein the one or more PRS measurements are performed without a measurement gap.

20. The apparatus of claim **15**, wherein the processor circuitry is further to:

decode a message that activates a measurement gap for the one or more PRS measurements; and

use the measurement gap upon expiration of an activation delay time.

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